Lecture 2 – comparison, limit comparison & Ratio test

Eg: $\sum \frac{1}{n^2} \to$ conv. by p series test. How about $\sum \frac{1}{(n+1)^2}$ or $\sum \frac{1}{n(n+1)}$?

Comparison test: If $\sum a_n$ & $\sum b_n$ are +ve term series then if $a_n \le b_n$ $\forall n \ge m$, $m \in \mathbb{N}$. Then 1) If $\sum b_n$ is cgt then $\sum a_n$ is cgt.

2) If $\sum a_n$ is div then $\sum b_n$ is div.

Eg 1) $\sum \frac{1}{\sqrt{5n^2+7n+11}}$ $\because 5n^2+7n+11 \le 23n^2 \Rightarrow \frac{1}{\sqrt{5n^2+7n+1}} \ge \frac{1}{n\sqrt{23}}$

& $\sum \frac{1}{n\sqrt{23}}$ is div. Hence by comparison test $\sum \frac{1}{\sqrt{5n^2+7n+11}}$ is div.

2) $\sum \frac{1}{n!}$ $\because 2^{n-1} \le n!$ $\forall n \ge 1$ $\Rightarrow$ $\frac{1}{n!} \le \frac{1}{2^{n-1}}$ & $\sum \frac{1}{2^{n-1}}$ conv. (G.P.)

Hence by comp. test $\sum \frac{1}{n!}$ is conv.

(discussed in previous class) | OK 2

Limit comparison test:

1) If $\sum a_n$ & $\sum b_n$ are +ve term series s.t. $\lim_{n\to\infty} \frac{a_n}{b_n} \ne 0$ (but finite) then $\sum a_n$ & $\sum b_n$ both conv. or diverge together

2) If $\lim \frac{a_n}{b_n} = 0$. Then if $\sum b_n$ is cgt then $\sum a_n$ is also cgt & if $\sum a_n$ is div then $\sum b_n$ is also div.

3) If $\lim \frac{a_n}{b_n} = \infty \Rightarrow \lim \frac{b_n}{a_n} = 0$ & apply 2).

Eg 1) $\sum \frac{1}{\sqrt{5n^2+7n+11}}$ Take $b_n = \frac{1}{n}$ then $\lim \frac{a_n}{b_n} = \lim \frac{n}{\sqrt{5n^2+7n+11}} = \frac{1}{\sqrt{5}}$ $\ne 0$ & finite

then $\because \sum b_n$ is div, $\sum a_n = \sum \frac{1}{\sqrt{5n^2+7n+1}}$ is div. also

2) $\sum \frac{1}{n^2 e^n}$ Take $b_n = \frac{1}{n^2}$ then $\lim \frac{a_n}{b_n} = \lim \frac{1}{e^n} = 0$

& $\sum b_n$ is cgt (by p-series test) $\therefore \sum a_n = \sum \frac{1}{n^2 e^n}$ is cgt.

3) $\sum \frac{2^n}{n^3}$ If you take $b_n = \frac{1}{n^3}$ then $\lim \frac{a_n}{b_n} \to \infty$

but $\sum b_n$ is cgt (for application of limit comparison test, it should have been div.)

Ratio test: If $\lim \frac{a_{n+1}}{a_n} = l$

$l > 1$ div

$l < 1$ cgt

$l = 1$ test fails.

Eg. 1) $\sum \frac{3^n}{n^3}$ $\quad \lim \frac{a_{n+1}}{a_n} = \lim \frac{3^{n+1}}{(n+1)^3} \cdot \frac{n^3}{3^n} = 3 \cdot \lim \frac{n^3}{(n+1)^3} = 3 > 1$

2) $\sum \frac{n!}{n^n}$ $\quad \lim \frac{a_{n+1}}{a_n} = \lim \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \lim \frac{n^n}{(n+1)^n}$ (div.)

$= \lim \frac{1}{\left(1+\frac{1}{n}\right)^n} = \frac{1}{e} < 1$ Hence cgt. $\because \lim_{n\to\infty} [1+f(n)]^{g(n)} = e^{f(n)\cdot g(n)}$

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